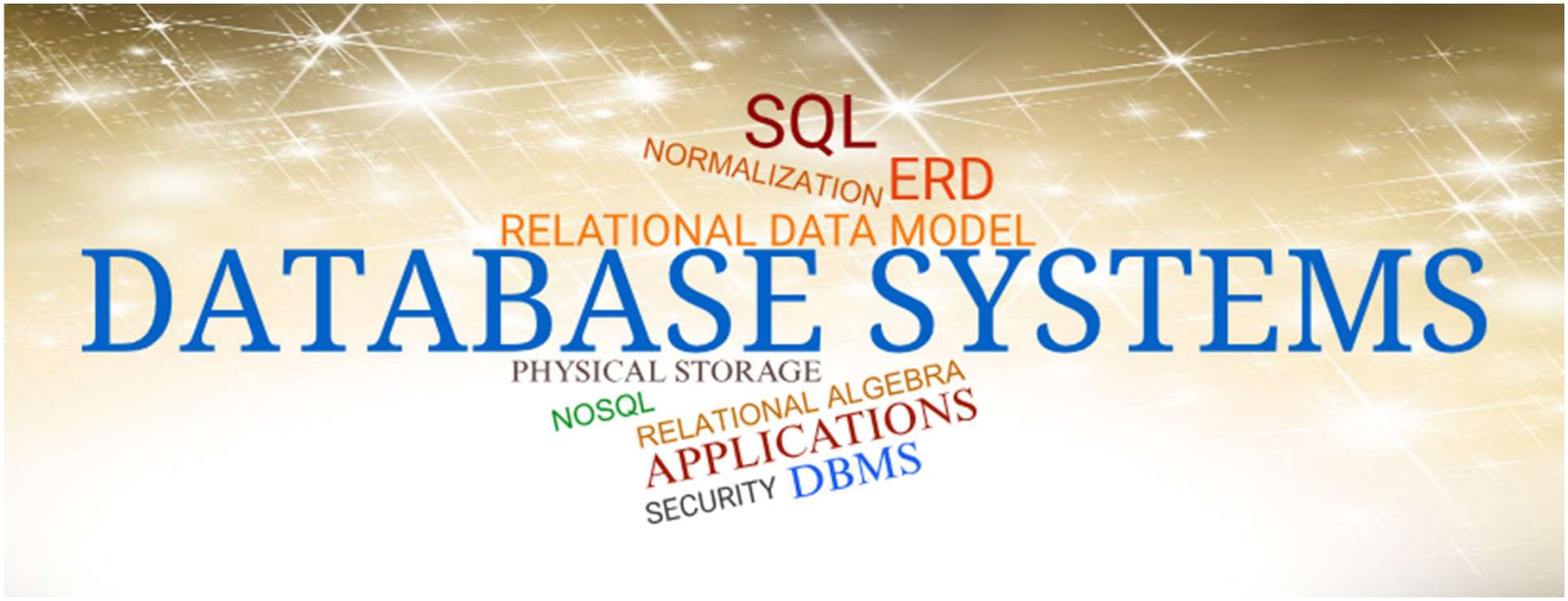


Database Systems

Overview

Chapter 1

Contents

1	Basic concepts
2	File-based Approach
3	Database Approach
4	Data Models, Schemas, and Instances
5	Three-Schema Architecture and Data Independence
6	Database Languages
7	Emerging Database Applications

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Basic concepts

- Data, information, knowledge and wisdom
- Database
- Database management system (DBMS)
- Application program
- Types of database applications

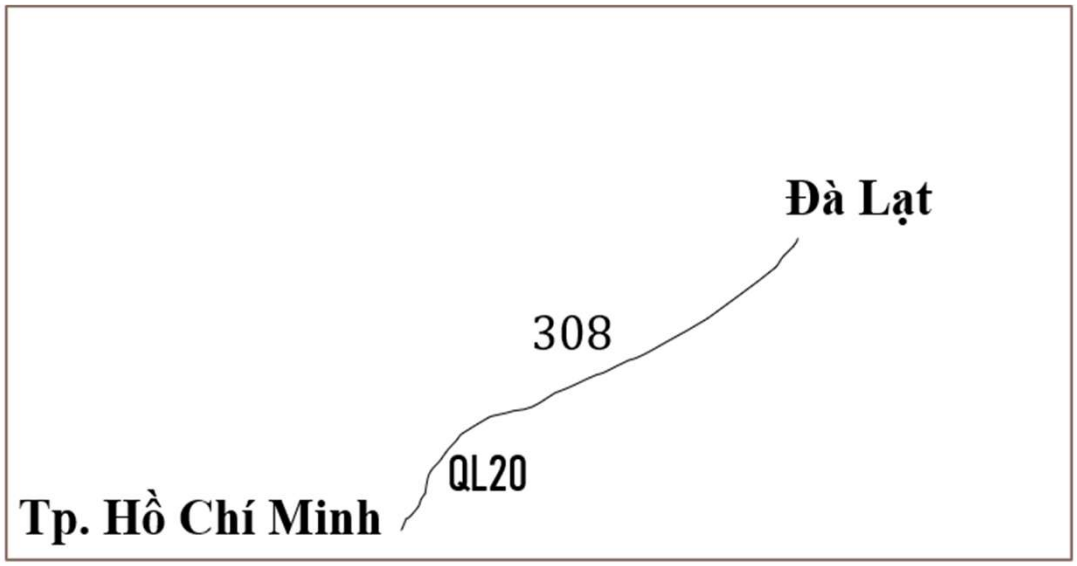
Basic concepts

- **Data**
  - Known facts that can be recorded and that have implicit meaning.
- **Information**
  - Result of analyzing and interpreting pieces of data.
- **Knowledge**
  - Information combined with experience, understanding, and insights to draw conclusions.
- **Wisdom**
  - Judgments and decisions using knowledge, experience, and ethical considerations.

Basic concepts

- **Data** (raw facts): *308*
  Just a number

- **Information** (processed data with context): *308 km*
  "The distance from Ho Chi Minh City to Da Lat is about 308 km via QL20"



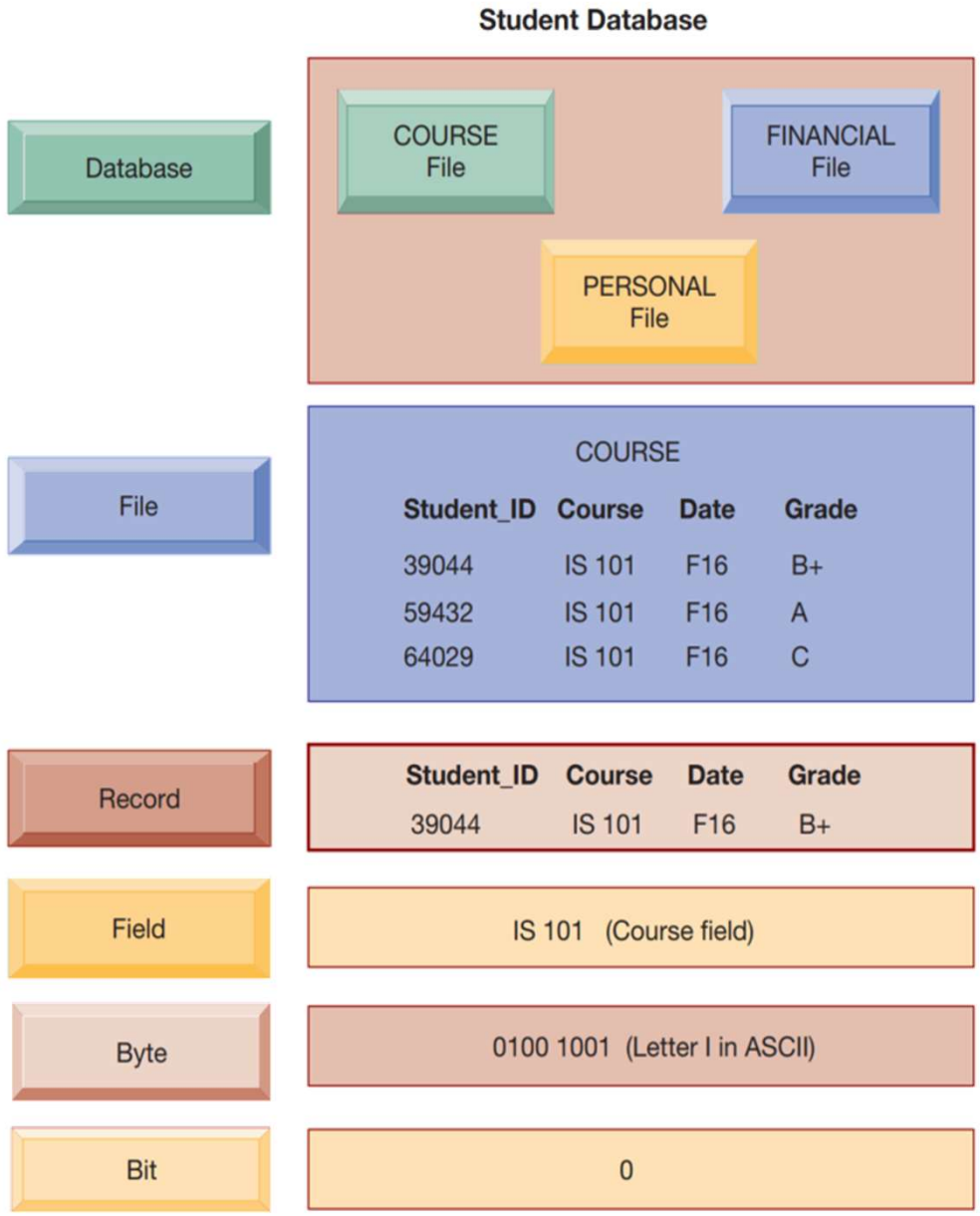
Đvt: km

- **Knowledge** (understanding): *"308 km is a quite far distance"*

- **Wisdom** (judgment & action): *"It's very difficult to walk 308 km by any person, but vehicle transport is OK"*

Basic Concepts

- ***Database***: group of related files.
- ***File***: group of records of the same type
- ***Record***: group of related fields
- ***Field***: group of characters into word(s), or a number
- ***Byte***: group of bits, represents a single character, which can be a letter, a number, or another symbol.
- ***Bit*** represents the smallest unit of data a computer can handle.



Database



- *A collection of related data with an implicit meaning*
- Implicit properties:
  - Represents some aspect of the real world (*miniworld*)
  - A *logically coherent* collection of with some inherent meaning.
  - Designed, built, and populated with data for a *specific purpose*.

- **UNIVERSITY** database
  - Information concerning students, lecturers, courses, and grades in a university environment.

Example: a part of University Database

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

Database

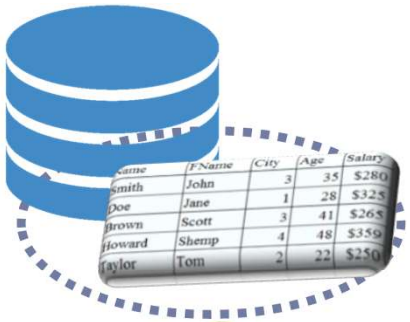
- **Database *manipulation*** involves querying and updating.
- Examples of queries are as follows:
  - Retrieve the transcript—a list of all courses and grades—of 'Smith'
  - List the names of students who took the section of the 'Database' course offered in fall 2008 and their grades in that section
  - List the prerequisites of the 'Database' course
- Examples of updates include the following:
  - Change the class of 'Smith' to sophomore
  - Create a new section for the 'Database' course for this semester
  - Enter a grade of 'A' for 'Smith' in the 'Database' section of last semester

Basic concepts

- **Database management system (DBMS):** A computerized system to create and maintain a database.
- An **application program** accesses the database by sending queries or requests for data to the DBMS.
  - A *query* typically causes some data to be retrieved
  - A *transaction* may cause some data to be read and some data to be written into the database.



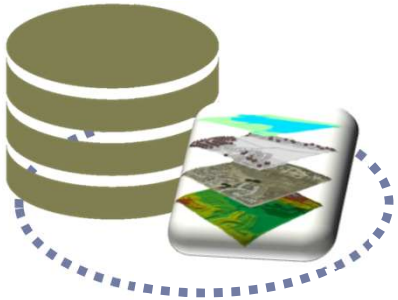
Some of database applications



- **Traditional database**: textual or numeric information



- **Multimedia database**: images, audio clips, and video streams



- **Spatial database**: geometric objects
- **Geographic information systems (GIS)**: store and analyze maps, weather data, and satellite images



- **Temporal database**: historical data

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File-based Approach

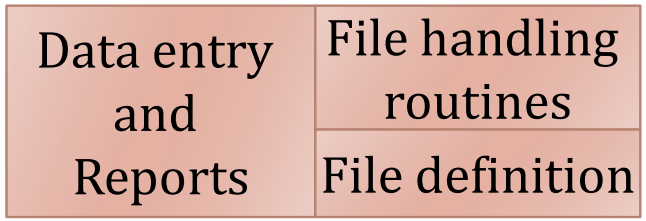
- Data is stored in one or more separate computer files
- Data is then processed by computer programs - applications

File-based Approach



Sales Department





Sales Application Programs

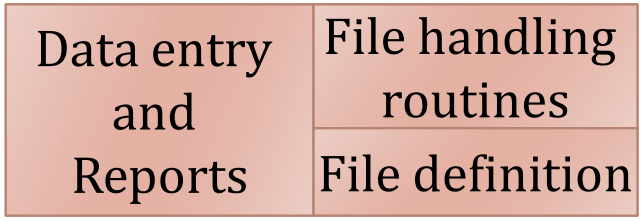






Contract Department





Contract Application Programs





Sales files

PrivateOwner (ownerNo, fName, lName, address, telNo)
PropertyForRent (propertyNo, street, postcode, rooms, ownerNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Lease (leaseNo, propertyNo, clientNo, deposit, paid, Start, Finish)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

Contract files

File-based Approach

- **Problems:**
  - ***Data Redundancy***
    - The same information being kept in several different places (files)
    - Wastes storage space and duplicates effort
  - ***Data Inconsistency***
    - Various copies of the same data are conflicting
    - Inconsistency in data format

File-based Approach

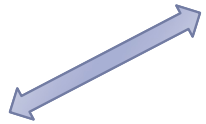


Sales Department



Data entry and Reports	File handling routines
	File definition

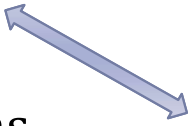
Sales Application Programs



PropertyForRent File



PrivateOwner File



Client File

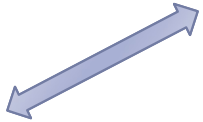


Contract Department



Data entry and Reports	File handling routines
	File definition

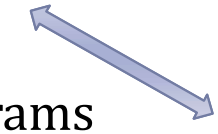
Contract Application Programs



Lease File



PropertyForRent File



Client File

Shared File Approach



Sales Department

Data entry and Reports	File handling routines
	File definition

Sales Application Programs



Contract Department

Data entry and Reports	File handling routines
	File definition

Contract Application Programs

PrivateOwner File

PropertyForRent File

Client File

Lease File

Shared File Approach

- Data (files) is shared between different applications
- Data redundancy problem is alleviated
- Data inconsistency problem across different versions of the same file is solved

Shared File Approach

- Problems:
  - ***Rigid data structure***: If applications have to share files, the file structure that suits one application might not suit another
  - ***Physical data dependency***: If the structure of the data file needs to be changed in some way, this alteration will need to be reflected in all application programs that use that data file
  - ***No support of concurrency control***: While a data file is being processed by one application, the file will not be available for other applications or for ad hoc queries

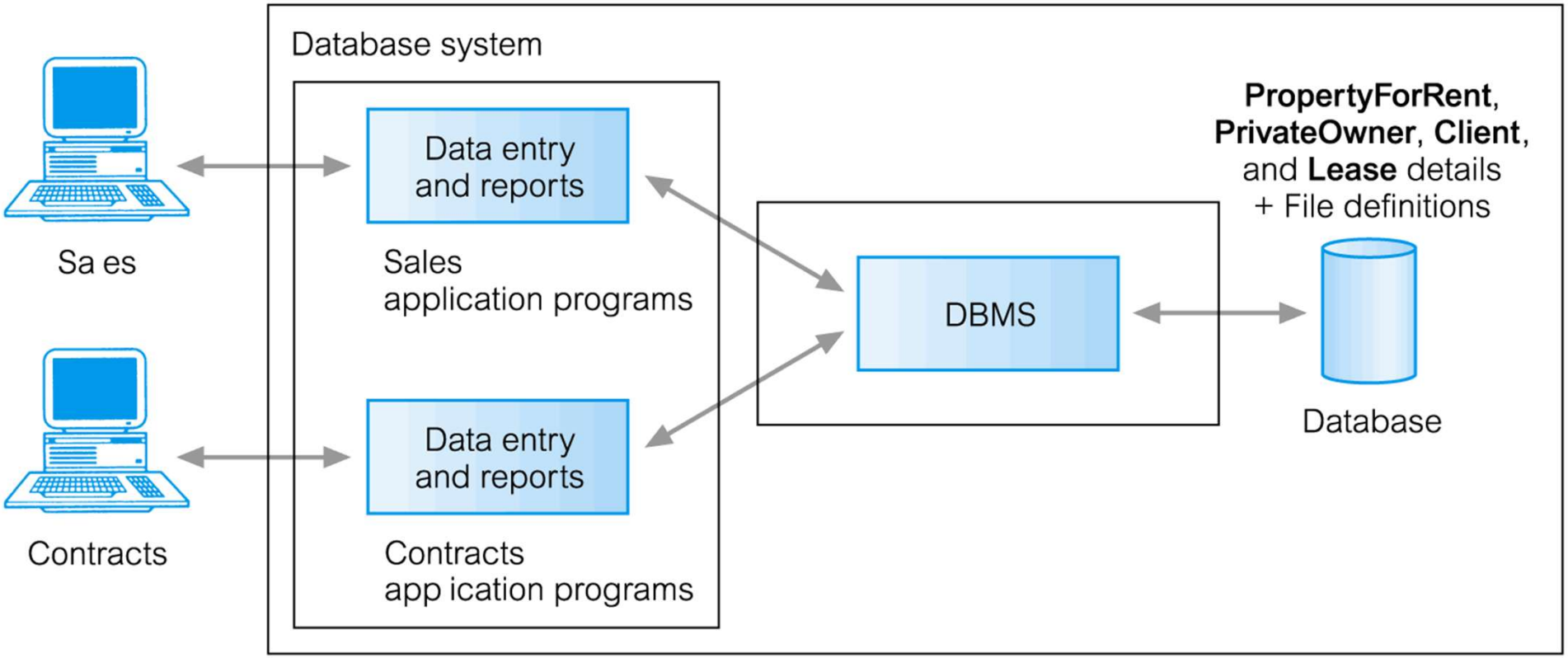
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Database Approach

- Database approach allows user:
  - Specify data types, structures and any data constraints to be stored in the database. All specifications are stored in the database
  - Query data: retrieve (query), update (insert, delete, modify)
  - Control access to database:
    - a security system
    - an integrity system
    - a concurrency control system
    - a recovery control system
    - a user-accessible catalog
- **Database System** = Database + DBMS software

Database Approach



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentF nish)

Database Approach

- System catalog (metadata) provides description of data to enable program–data independence
- Logically related data comprises *entities, attributes*, and *relationships* of an organization's information
- **D**ata**B**ase **M**anagement **S**ystem (DBMS):
  - A computerized system to create and maintain a database.
  - A general-purpose software system that facilitates the processes of defining, constructing, manipulating, and sharing databases among various users and applications

Database Catalog of University Database

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....	....	.....
....	....	.....
....	....	.....
Prerequisite_number	XXXXNNNN	PREREQUISITE

Functions of Database Management System

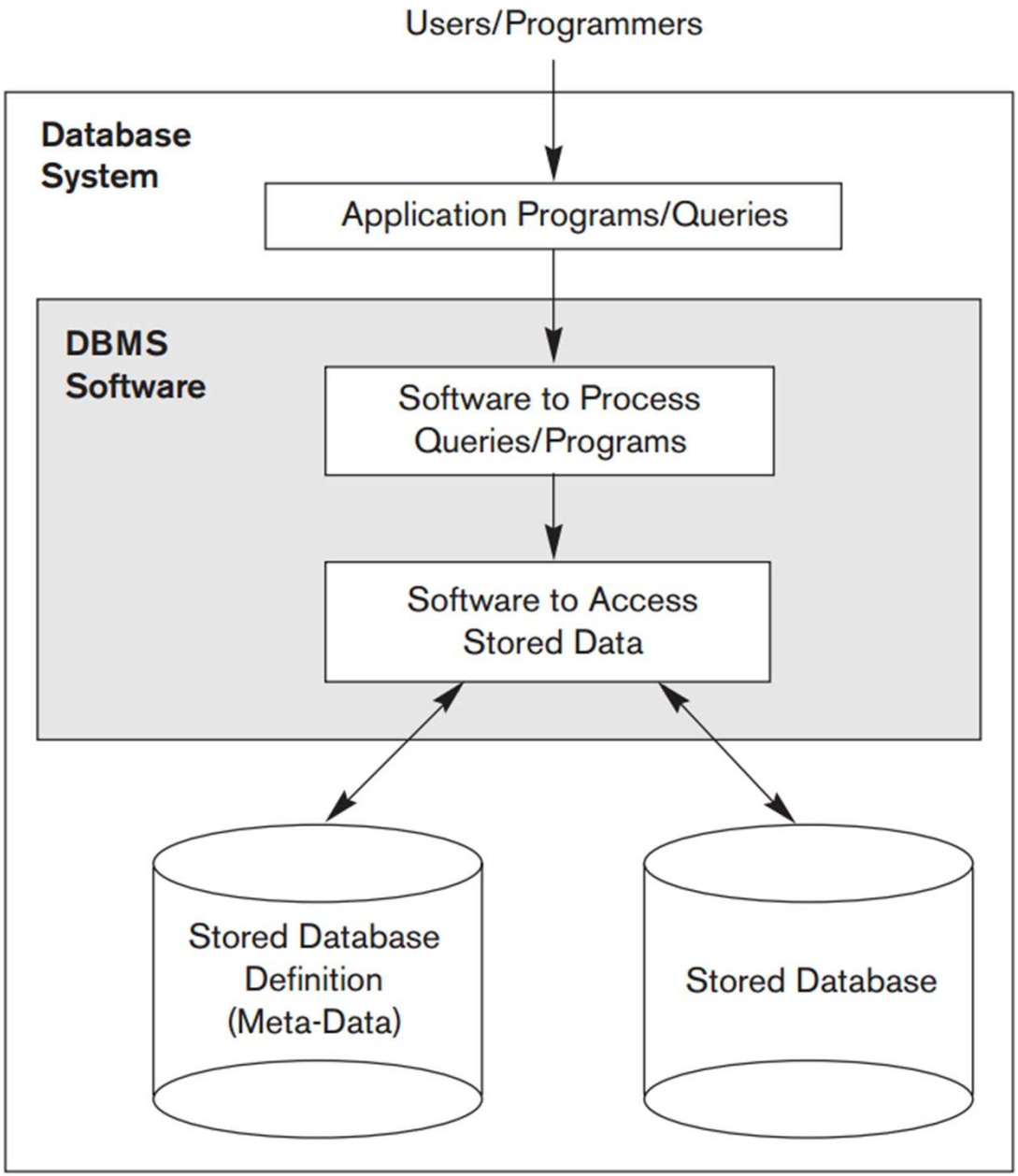
- ***Defining*** a database: specifying the data types, structures, and constraints for the data to be stored in the database.
- ***Constructing*** a database: storing the data itself on some storage medium that is controlled by the DBMS.
- ***Manipulating*** a database: querying the database to retrieve specific data, updating the database to reflect changes in the miniworld, and generating reports from the data.
- ***Sharing*** a database: allowing multiple users and programs to access the database concurrently.
- ***Protection*** includes both *system protection* against hardware or software malfunction (or crashes), and *security protection* against unauthorized or malicious access.
- ***Maintaining*** a database system: allowing the system to evolve as requirements change over time.

Characteristics of the Database Approach

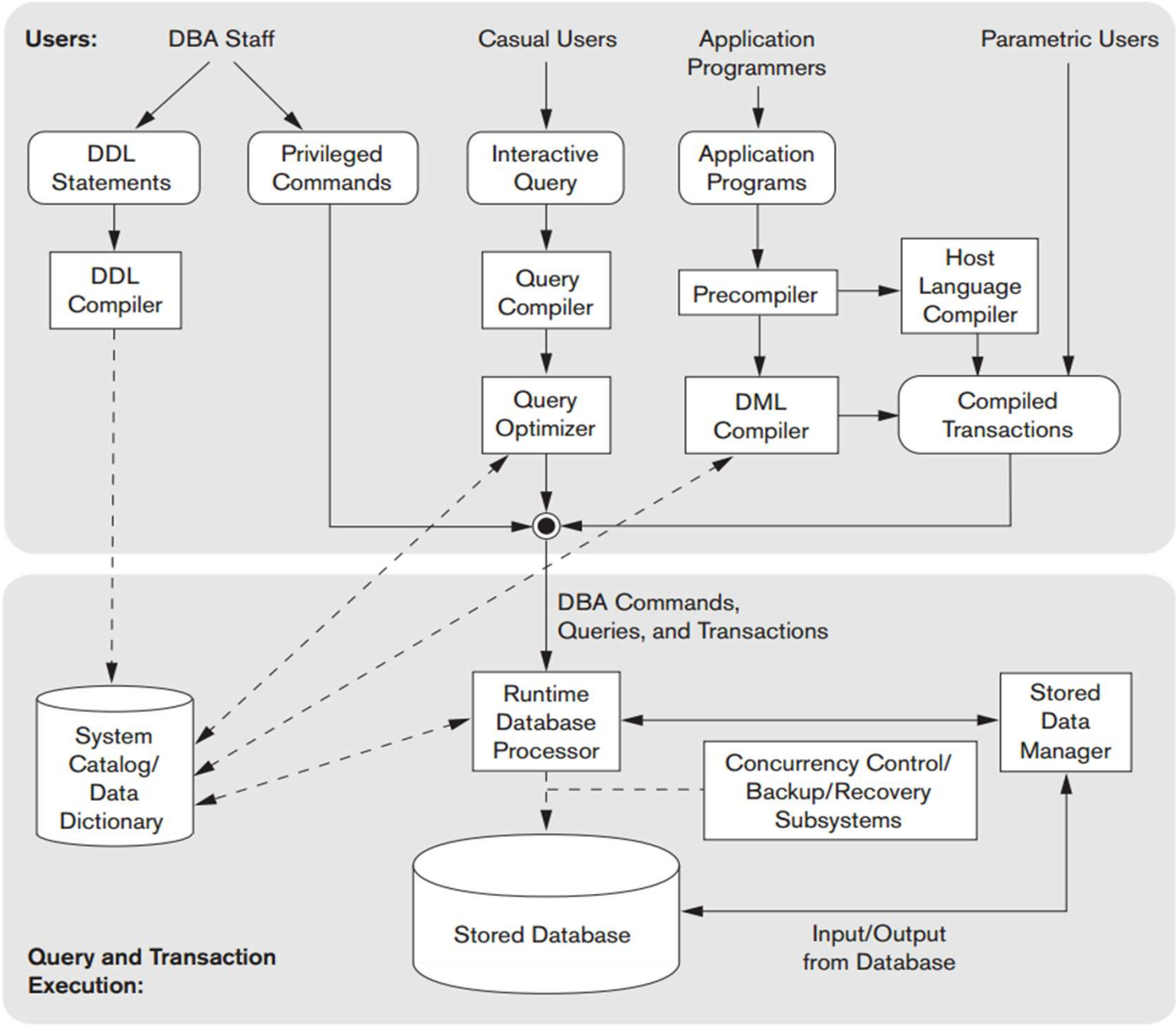
- Self-describing nature of a database system
  - The database system contains not only database itself but also a complete definition of the database structure and constraints (*catalog* or *meta-data*)
- Insulation between programs and data, and data abstraction
  - Program-data independence + Program-operation independence = Data abstraction
- Support of multiple views of the data
- Sharing of data and multi-user transaction processing

Reading note 1

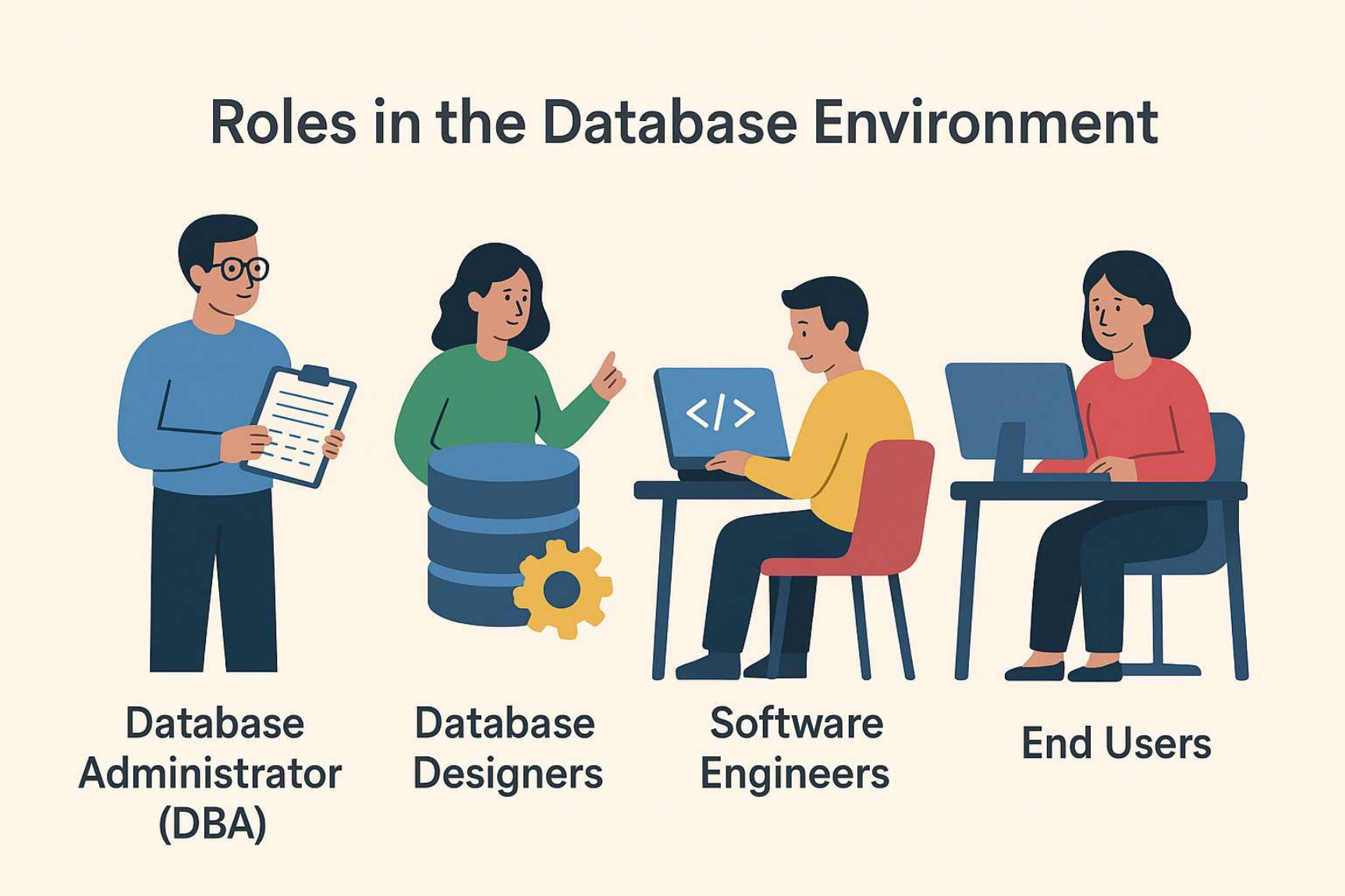
A Simplified Database System Environment



DBMS components



Roles in the Database Environment



Roles in the Database Environment

- **Database administrators (DBA)**:
  - Authorizing access to the database
  - Coordinating and monitoring its use
  - Acquiring software and hardware resources
- **Database designers**:
  - Identifying the data to be stored
  - Choosing appropriate structures to represent and store this data
- **Software Engineers**
  - System analysts: determining specifications of end users
  - Application programmers: implementing these specifications as programs.
- **End users**
  - People whose jobs require access to the database through the applications

Roles in the Database Environment

- **Workers behind the Scene**
  - **DBMS system designers and implementers**
    - Design and implement the DBMS modules and interfaces as a software package
  - **Tool developers**
    - Design and implement **tools**
  - **Operators and maintenance personnel**
    - Responsible for running and maintenance of hardware and software environment for database system

Classification of Database Management Systems

Several criteria can be used to classify DBMSs

- ***Data models*** (widely-used)
  - Relational data model (SQL system), Object-oriented data model, Object relational data model, XML-enabled data model, XML native data model, Graph data model, ...
- ***Number of users***:
  - Single-user, multiuser
- ***Architectures***:
  - Centralized, distributed, parallel
- ***Cost***:
  - Open source, licenses

Advantages of Using the DBMS Approach

- Controlling Redundancy
- Restricting Unauthorized Access
- Providing Persistent Storage for Program Objects
- Providing Storage Structures and Search Techniques for Efficient Query Processing
- Providing Backup and Recovery
- Providing Multiple User Interfaces
- Representing Complex Relationships among Data
- Enforcing Integrity Constraints
- Permitting Inferencing and Actions using Rules and Triggers

When NOT to Use a DBMS

- Overhead costs of using a DBMS:
  - High initial investment in hardware, software, and training
  - The generality that a DBMS provides for defining and processing data
  - Overhead for providing security, concurrency control, recovery, and integrity functions
- More desirable to use regular files for:
  - Simple, well-defined database applications not expected to change at all
  - Stringent, real-time requirements that may not be met because of DBMS overhead
  - Embedded systems with limited storage capacity
  - No multiple-user access to data



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Data Model

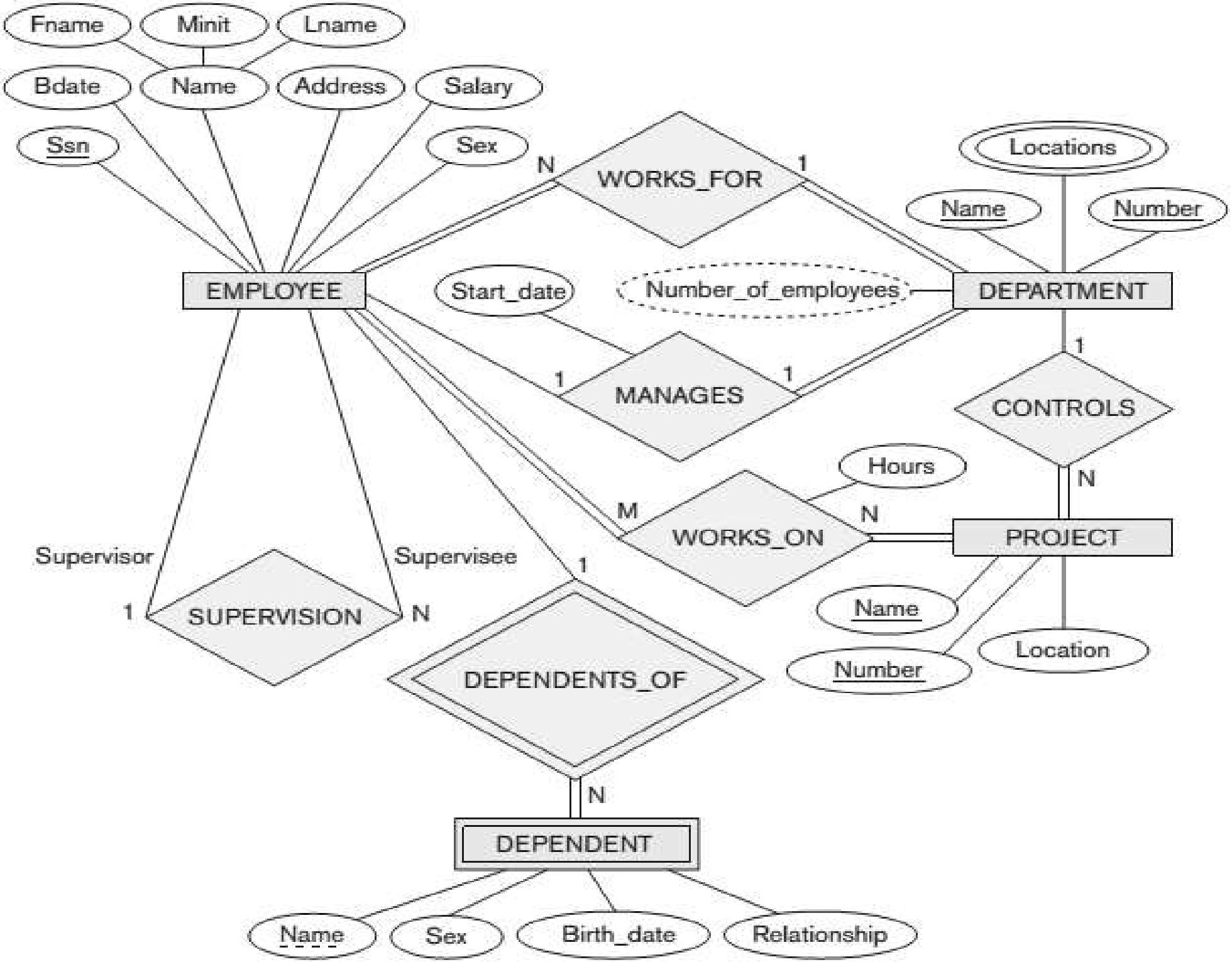
- Data model: a collection of concepts that can be used to describe the structure of a database.
  - Structure of a database: data types, relationships, and constraints that apply to the data.
  - Data models also include a set of basic operations for specifying retrievals and updates on the database.
- A data model is a type of data abstraction.

Categories of data models

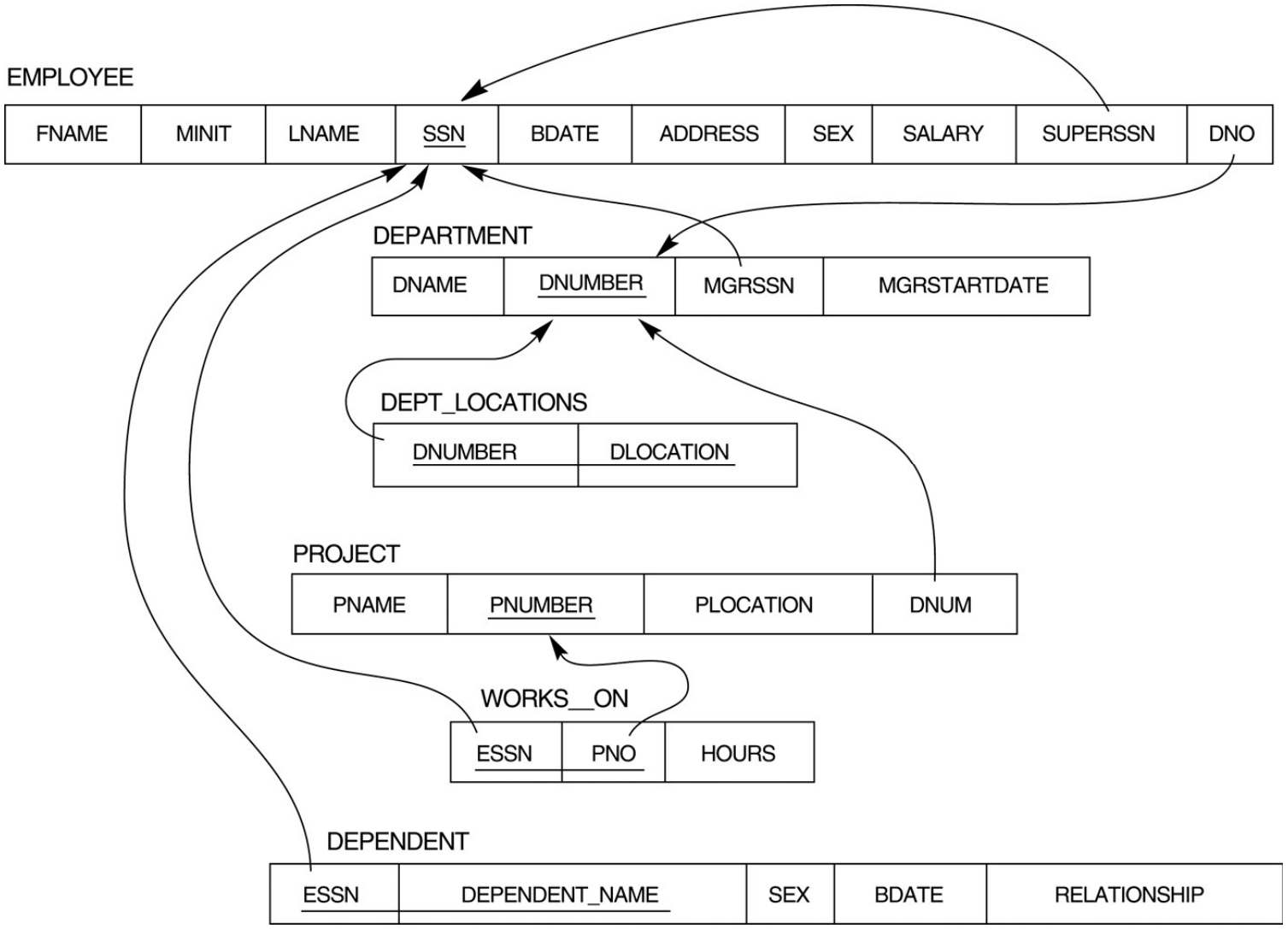
- ***High-level or conceptual data models***: provide concepts that are close to the way many users perceive data
  - Entity relationship model, object-oriented data model
- ***Representational or implementation data models***: provide concepts that are easy for end users to understand while still being close to how data is actually stored in a computer..
  - Relational data model
- ***Low-level or physical data models***: provide concepts that describe the details of how data is stored in the computer.

Database Schemas and Instances

- ***Database Schema***: the *description* of a database, which is specified during database design and is not expected to change frequently
- *Schema Diagram*: a displayed schema
- ***Database Instance*** (database state, snapshot): the *data* in the database at a particular moment in time


Fname
Minit
Lname
Name
Bdate
Address
Salary
Ssn
Sex
N
WORKS_FOR
1
Locations
Name
Number
EMPLOYEE
Start_date
Number_of_employees
DEPARTMENT
1
1
MANAGES
1
CONTROLS
Hours
N
M
N
WORKS_ON
PROJECT
Supervisor
Supervisee
1
1
SUPERVISION
N
Name
Location
Number
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
Birth_date
Relationship

Relational Database Schema Diagram



Database state

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

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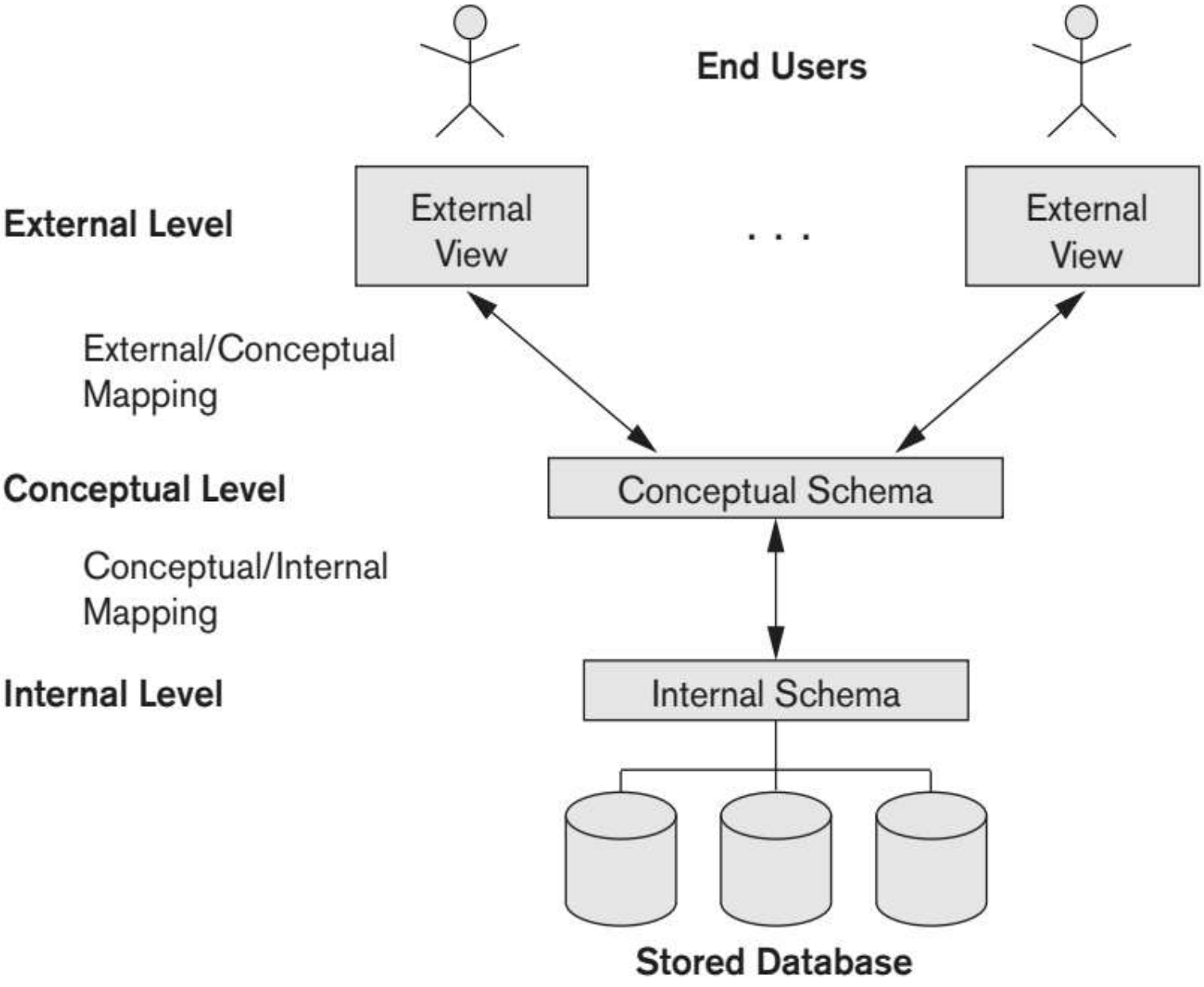
4 Data Models, Schemas, and Instances

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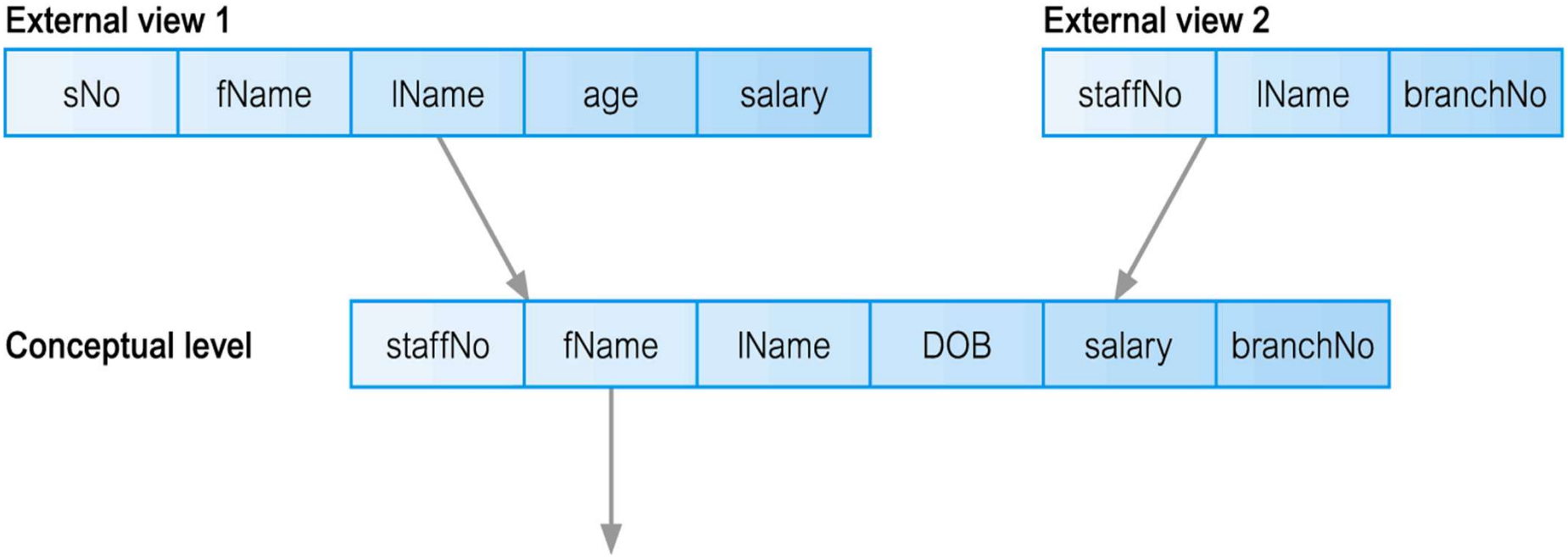
Three-Schema Architecture and Data Independence



Three-Schema Architecture and Data Independence

- **External Level**
  - Users' view of the database
  - Describes that part of database that is relevant to a particular user
- **Conceptual Level**
  - Community view of the database
  - Describes what data is stored in database and relationships among the data
- **Internal Level**
  - Physical representation of the database on the computer.
  - Describes how the data is stored in the database

Three-Schema Architecture and Data Independence



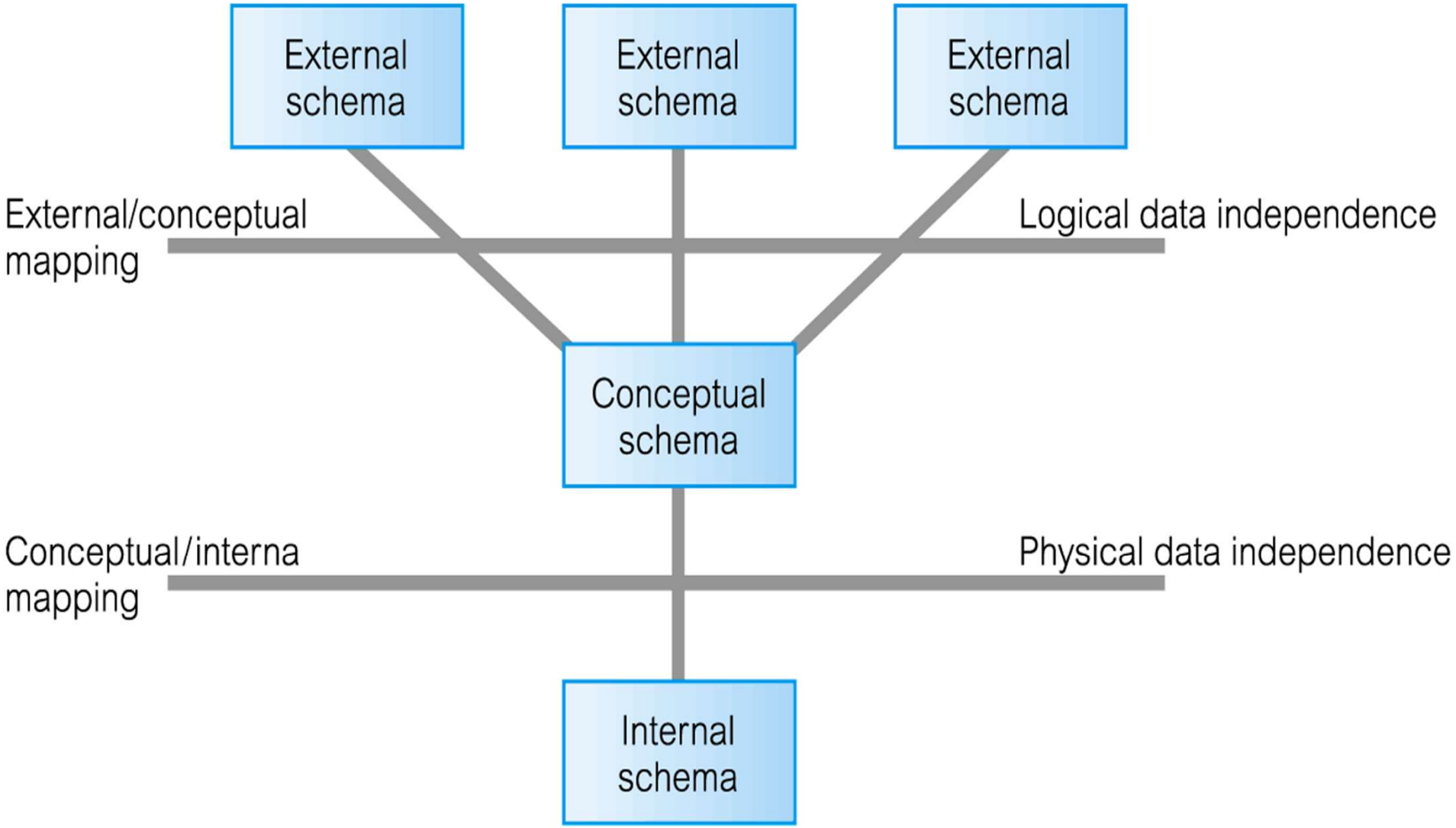
Internal level

```
struct STAFF {
        int staffNo;
        int branchNo;
        char fName [15];
        char lName [15];
        struct date dateOfBirth;
        float salary;
        struct STAFF *next;                 /* pointer to next Staff record */
};
index staffNo; index branchNo;             /* define indexes for staff */
```

Three-Schema Architecture and Data Independence

- **Data Independence:** is the capacity to change the schema at one level of a database system without having to change the schema at the next higher levels
- Logical Data Independence:
  - Conceptual schema changes (e.g. addition/removal of entities) should not require changes to external schema or rewrites of application programs
- Physical Data Independence:
  - Internal schema changes (e.g. using different file organizations, storage structures/devices) should not require changes to conceptual or external schemas

Three-Schema Architecture and Data Independence



Three-Schema Architecture and Data Independence

- Objectives of Three-Schema Architecture
  - All users should be able to access same data
  - Users should not need to know physical database storage details
  - DBA should be able to change database storage structures without affecting the users' views. *Internal structure of database should be unaffected by changes to physical aspects of storage.*
  - DBA should be able to change conceptual structure of database without affecting all users

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Database Languages

- **Data Definition Language (DDL)** allows the DBA or user to describe and name entities, attributes, and relationships required for the application plus any associated integrity and security constraints
- **Data Manipulation Language (DML)** provides basic data manipulation operations (select, insert, update, delete) on data held in the database
- **Data Control Language (DCL)** defines activities that are not in the categories of those for the DDL and DML, such as granting privileges to users, and defining when proposed changes to a databases should be irrevocably made

Database Languages

- Procedural DML allows user to tell system exactly how to manipulate data (e.g., Network and hierarchical DMLs)
- Non-Procedural DML (declarative language) allows user to state what data is needed rather than how it is to be retrieved (e.g., **SQL**, QBE)
- Fourth Generation Languages (4GLs)
  - Non-procedural languages: SQL, QBE, etc.
  - Application generators, report generators, etc.

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Database Management Systems Framework

Application Layer	Visualization, Collaborative Computing, Mobile Computing, Knowledge-based Systems
Data Management Layer	**Layer 3: information extraction & sharing** Data Warehousing, Data Mining, Internet DBs, Collaborative, P2P & Grid Data Management
	Layer 2: interoperability & migration Heterogeneous DB Systems, Client/Server DBs, Multimedia DB Systems, Migrating Legacy DBs
	Layer 1: DB technologies **DB Systems**, Distributed DB Systems
Supporting Layer	Networking, Mass Storage, Agents, Grid Computing Infrastructure, Parallel & Distributed Processing, Distributed Object Management

Emerging Database Applications

- Emerging applications: storage and retrieval of images, videos, data mining (large amounts of data need to be stored and analyzed), spatial databases, time series applications, big data, cloud …
- More complex data structures than relational representation
- New data types except for the basic numeric and character string types
- New operations and query languages for new data types
- New storage and retrieval methods
- New security mechanisms
- …

Emerging Database Applications

- **Big data** storage systems, or **NOSQL systems:**
  - Manage data for social media applications.
- **Cloud storage**
  - Users are provided with storage capabilities on the Web.
- **Data warehouses and online analytical processing (OLAP) systems**:
  - Extract and analyze useful business information from very large databases
  - Support decision making
- **Real-time and active database technology**
  - Control industrial and manufacturing processes

Big data

1 Petabyte (PB) = 1,000 Terabytes (TB) = 1,000,000 Gigabytes (GB)
1 Exabyte (EB) = 1,000 Petabytes (PB)

- Traditional data: transaction records → rows & columns in relational DBMS.
- New sources:
  - Web traffic, emails, social media (tweets, posts).
  - Machine-generated data: smart meters, sensors, trading systems.
- Problem: unstructured/semi-structured → not suitable for relational DB.
- **What is Big Data?**
  - Extremely **large and complex datasets** beyond typical DBMS capabilities.
  - Not a fixed size → often **petabytes or exabytes** (billions–trillions of records).
  - Produced at **greater volume, variety, and velocity** than traditional data.

Data Warehouses & Data Marts

- **Data warehouse**:
  - Stores current and historical data from many core operational transaction systems
  - Consolidates and standardizes information for use across enterprise, but data cannot be altered
  - Provides analysis and reporting tools
- **Data marts**:
  - Subset of data warehouse
  - Summarized or focused portion of data for use by specific population of users
  - Typically focuses on single subject or line of business

Online analytical processing (OLAP)

- Supports multidimensional data analysis
  - Viewing data using multiple dimensions
  - Each aspect of information (product, pricing, cost, region, time period) is different dimension
  - Example: How many washers sold in the East in June compared with other regions?
- OLAP enables rapid, online answers to ad hoc queries

Data mining

- Finds hidden patterns, relationships in datasets
  - Example: customer buying patterns
- Infers rules to predict future behavior
- Types of information obtainable from data mining:
  - Associations
  - Sequences
  - Classification
  - Clustering
  - Forecasting

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